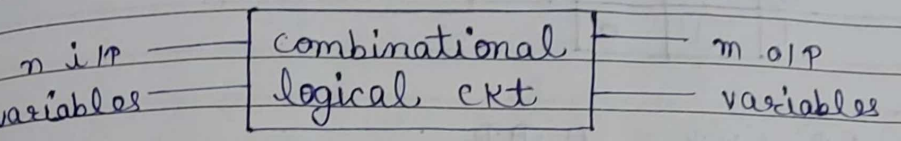
Page No.
Date

Unit - 3

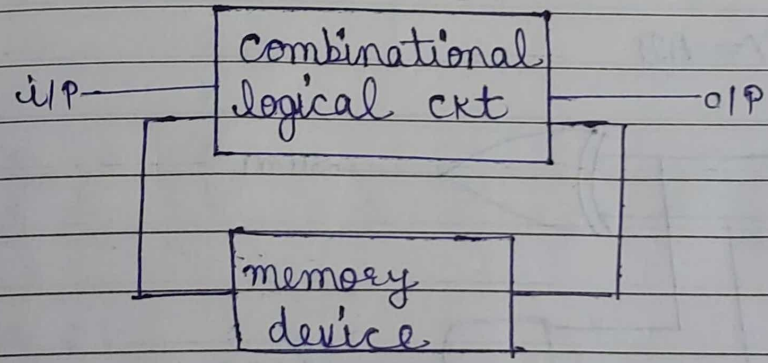
Combinational Logic circuit design

Combinational circuit :-



Sequential circuit :-

The values of o/p variables not only depends on the present i/p variables but also on the previous history of i/p, which are stored & retained in the memory device like flip-flop, resistors, counters etc.



Half adder :-

The ckt that adds two bits is called half adder. The truth table is given below.

i/p A	i/p B	o/p Sum	o/p Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Spiral

K-map for sum :-

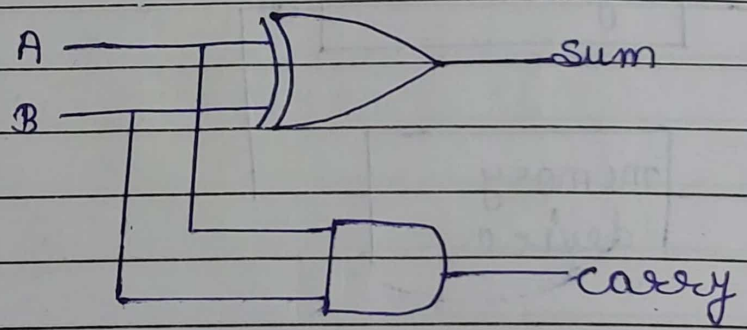
A \ B	0	1
0	0	(1)
1	(1)	0

$Y = A\bar{B} + \bar{A}B = A \oplus B$

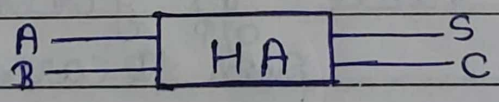
K-map for carry :-

A \ B	0	1
0	0	0
1	0	(1)

$Y = AB$



CKt diagram :-

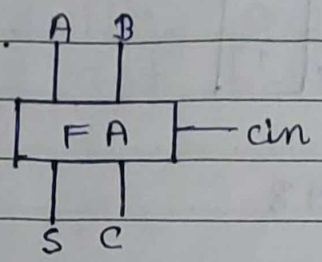


Full adder :-

A CKt that adds three bits is calle

Spiral

full adder. Here i/p are A, B, Cin.



Input			output	
A	B	cin	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K-map for S :-

A \ B cin	00	01	11	10
0	0	(1)	0	(1)
1	(1)	0	(1)	0

$Y = A\bar{B}\bar{C} + \bar{A}\bar{B}C + ABC + \bar{A}B\bar{C}$

$Y = \bar{C}(A\bar{B} + \bar{A}B) + C(\bar{A}\bar{B} + AB)$

$Y = \bar{C}(A \oplus B) + C(\overline{A \oplus B})$

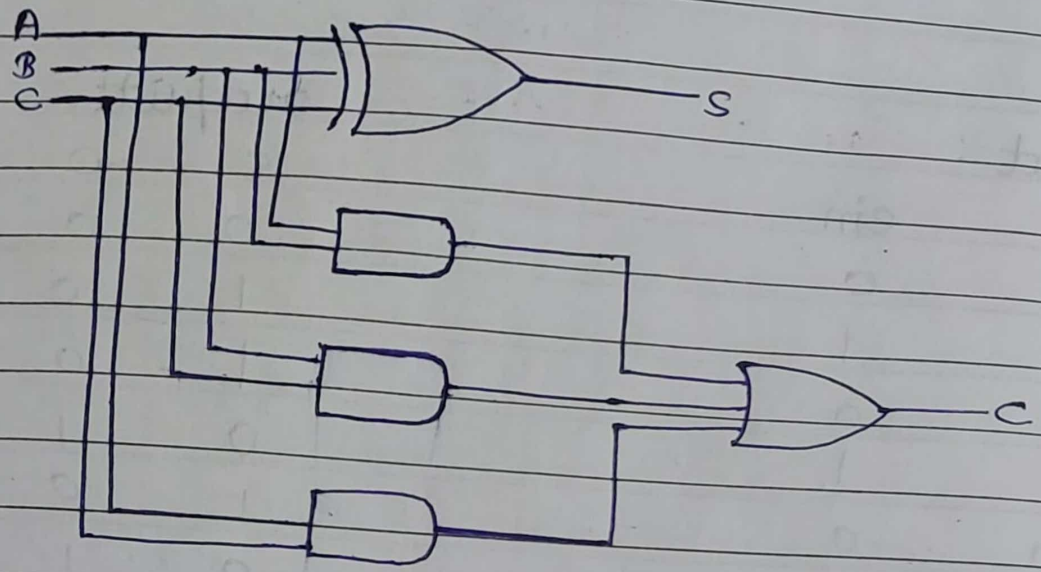
$Y = Cin \oplus A \oplus B$

Spiral

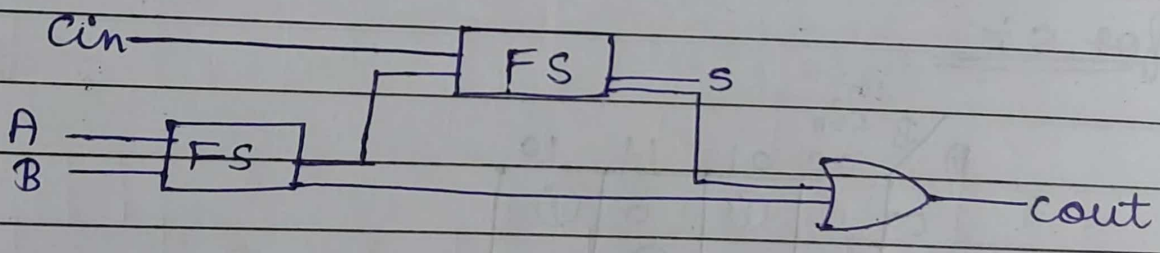
K-map for C ÷

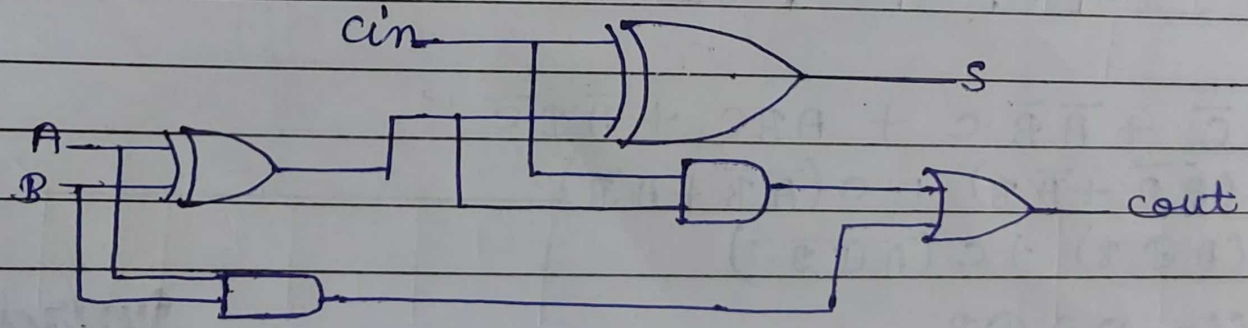
A \ BC	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$Cout = AC + BC + AB$$



Full adder using two half adder ÷





Page No.
Date

Half subtractor :-

It is a combinational ckt which subtract 2 bits and produced their difference D and borrow out (Bout)

Truth table :-

Input		output	
Miniend	subtractor	Diff.	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

K-map for diff.

X \ Y	0	1
0	0	1
1	1	0

K-map for Bout

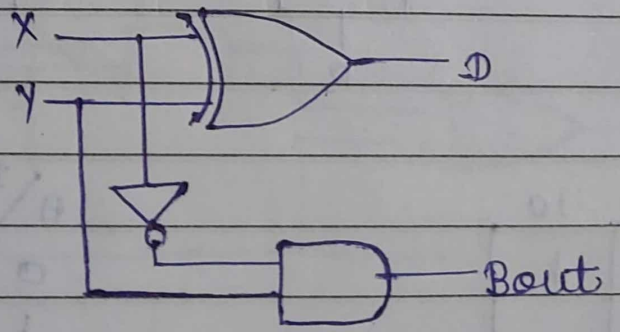
X \ Y	0	1
0	0	1
1	0	0

$D = \bar{X}Y + X\bar{Y}$

$D = X \oplus Y$

$Bout = \bar{X}Y$

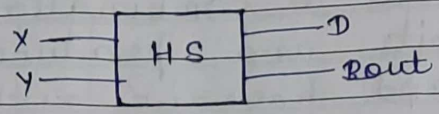
CKt diagram :-



Spiral

Page No.
Date

Block diagram :-

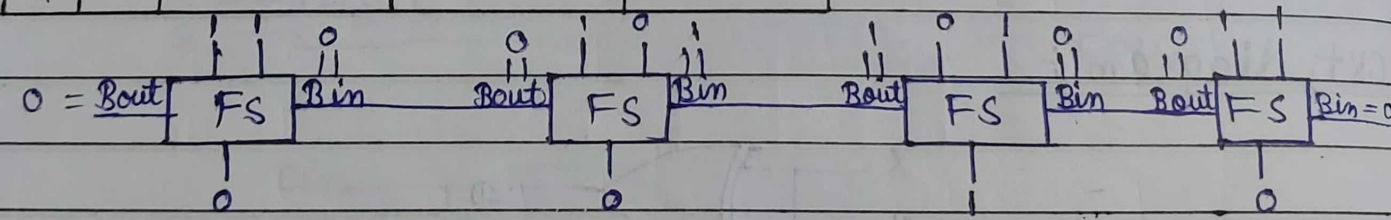


Full subtractor :-

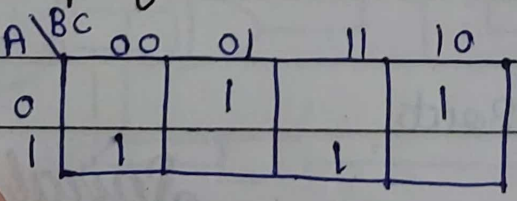
It has 3 i/p x, y & Bin, which denote minuend, subtrahend & borrow resp. The 2 o/p D & Bout represent the diff. & o/p borrow resp.

Truth table :-

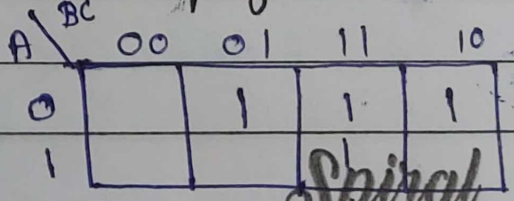
I/P			O/P	
A	B	Bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1



K-map for D



K-map for Bout



Spiral

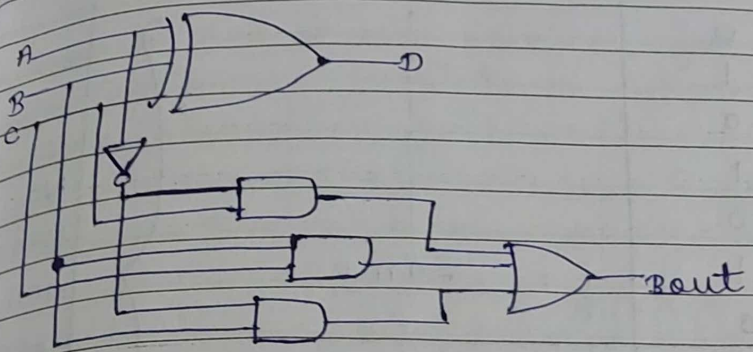
$D = A\bar{B}\bar{C} + \bar{A}\bar{B}C + ABC + \bar{A}B\bar{C}$

$= A(\bar{B}\bar{C} + BC) + \bar{A}(\bar{B}C + B\bar{C})$

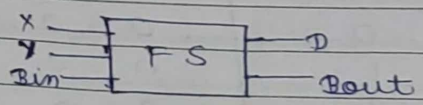
$= A(\bar{B} \oplus C) + \bar{A}(B \oplus C) \Rightarrow A \oplus B \oplus C$

$B_{out} = \bar{A}C + \bar{A}B + BC$

ckt diagram

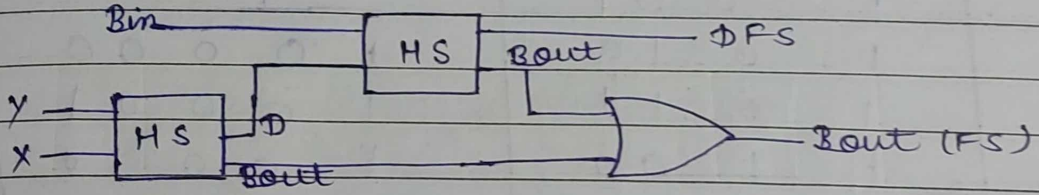


Block diagram

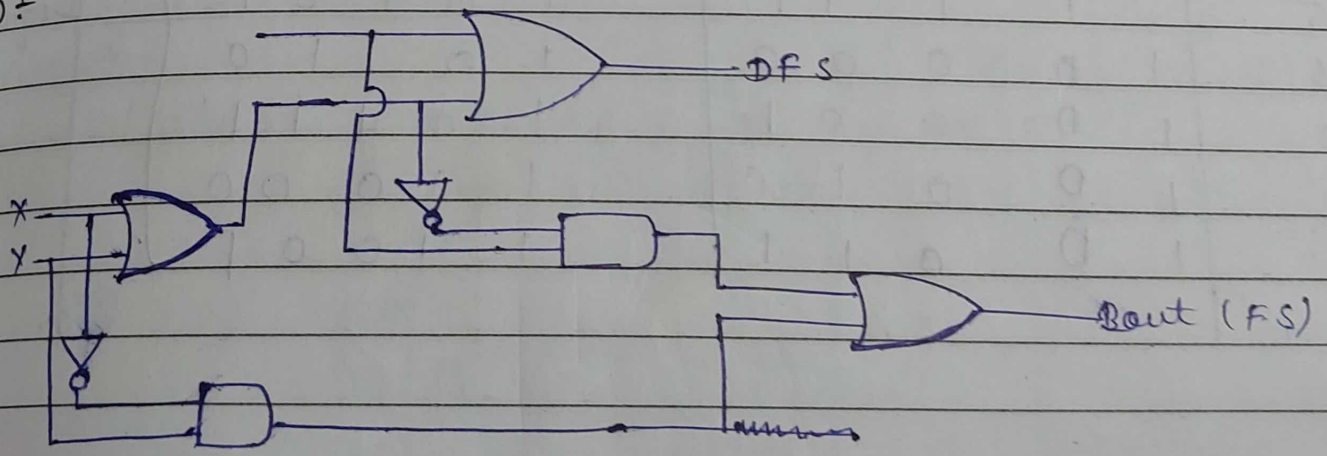


full subtractor using 2 half subtractor :-

BD :-



CD :-



Page No.
Date

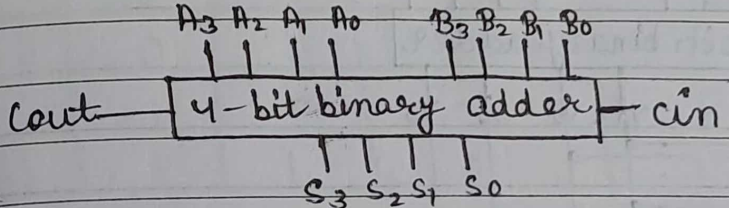
BCD adder

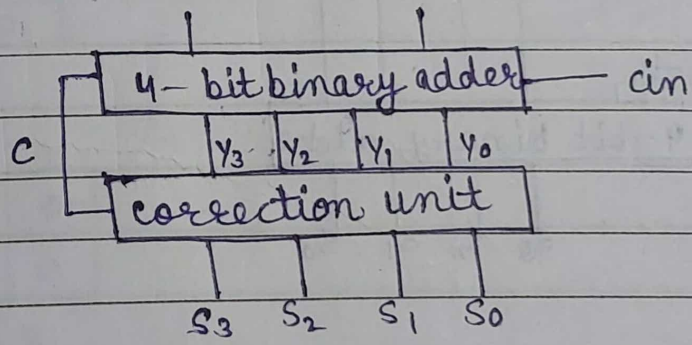
Decimal Digit	uncorrected BCD Sum C Y_3 Y_2 Y_1 Y_0	corrected BCD Sum Cout S_3 S_2 S_1 S_0	
0	0 0 0 0	0 0 0 0	No correction req
1	0 0 0 1	0 0 0 1	
2	0 0 1 0	0 0 1 0	
3	0 0 1 1	0 0 1 1	
4	0 1 0 0	0 1 0 0	
5	0 1 0 1	0 1 0 1	
6	0 1 1 0	0 1 1 0	
7	0 1 1 1	0 1 1 1	
8	1 0 0 0	1 0 0 0	
9	1 0 0 1	1 0 0 1	
10	1 0 1 0	1 0 0 0 0	Correction required
11	1 0 1 1	1 0 0 0 1	
12	1 1 0 0	1 0 0 1 0	
13	1 1 0 1	1 0 0 1 1	
14	1 1 1 0	1 0 1 0 0	
15	1 1 1 1	1 0 1 0 1	
16	1 0 0 0 0	1 0 1 1 0	
17	1 0 0 0 1	1 0 1 1 1	
18	1 0 0 1 0	1 1 0 0 0	
19	1 0 0 1 1	1 1 0 0 1	

• BCD adder is a combinational logic ckt that adds 2 BCD no. in parallel & produces a valid BCD no. as o/p.

• It consist of a :—

(a) 4 bit binary adder for intial addition.
(b) logic ckt to detect whether the sum is greater than 9 $(1001)_2$
(c) 4 bit adder to add 6 $(0110)_2$ which the sum if ~~suf~~ sum > 9 or carry c = 1.



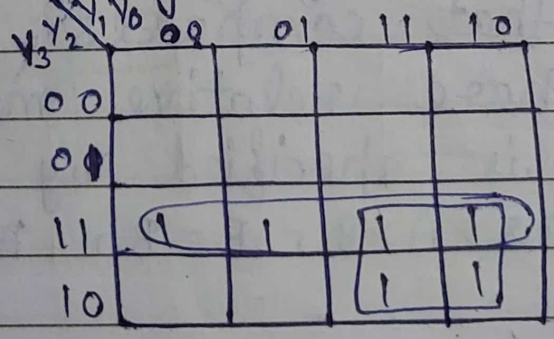


No correction req.

Correction required

Case ①

When range is for 10 to 15.



Spiral

$= Y_3 Y_2 + Y_3 Y_1'$

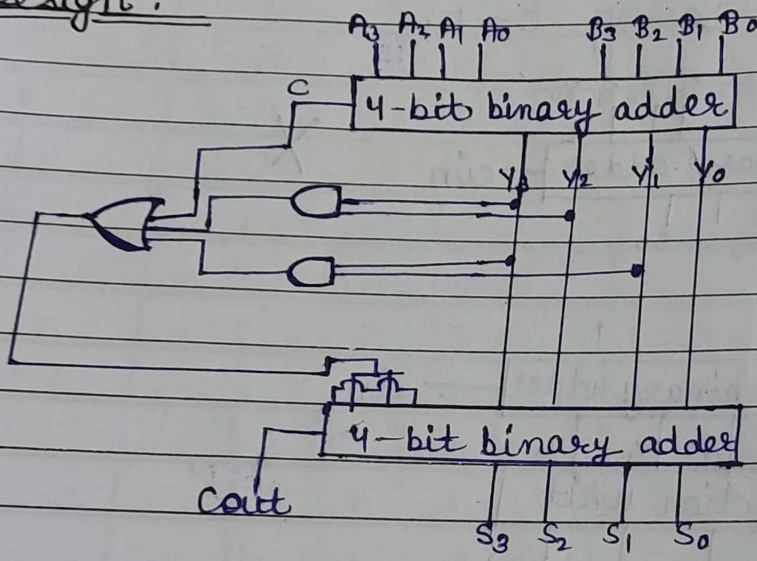
Case ②

When range is for 16 - 19.

In this case, we can signal the ckt when C = 1. So,

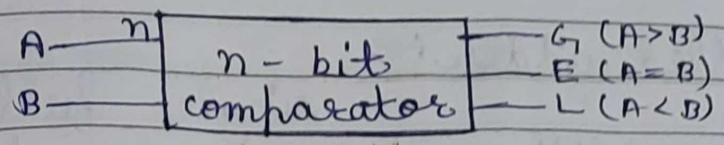
overall formula for adding 6 $= Y_3 Y_2 + Y_3 Y_1 + C$

Design :-



Magnitude Comparators :-

A mag. comparators is a combinational ckt that compares 2 numbers A, B & determine there relative magnitude. The outcome of comparators is specified by 3 binary variables that indicates A > B, A = B, A < B.



(i) 1-bit comparator :-

It is used to compare ~~2 bits~~ one bit of A & B

Truth table :-

A	B	G (A>B)	E (A=B)	L (A<B)
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

G :-

A \ B	0	1
0	0	0
1	1	0

$G = A \cdot \overline{B}$

E :-

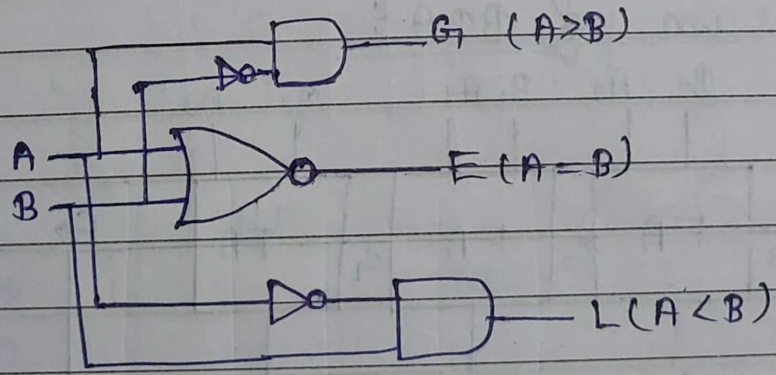
A \ B	0	1
0	1	0
1	0	1

$E = \overline{A}\,\overline{B} + AB \quad (A \odot B)$

L :-

A \ B	0	1
0	0	1
1	0	0

$L = \overline{A}B$



Spiral

Binary parallel adder :-

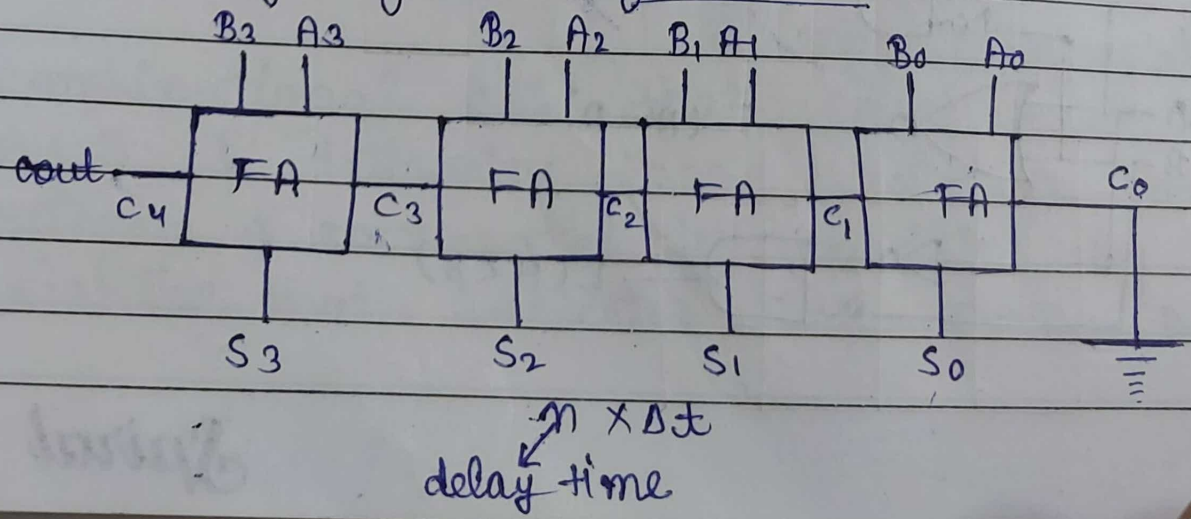
- A single full adder performs the addition of two one bit numbers and an i/p carry. But a parallel adder is a digital ckt capable of finding the arthmatic sum of 2 binary numbers i.e. greater than one bit in length by operating on corresponding pairs of bits in parallel.

- It consist of full adders connected in a chain, where the o/p carry from each full adder is connected to the carry i/p of next higher order full adder in the chain.

- A 'n' bit parallel adder requires 'n' full adder to perform the operation.

- So, for 2 bit no., two full adders are needed while for 4 bit no., 4 full adders are needed and so on - - - - -

Block dig diagram of BPA :-



Page No.
Date

Significant position 4th 3rd 2nd 1st

i/p carry — 1 1 1 0

Augend word A — A_3 A_2 0 A_1 A_0

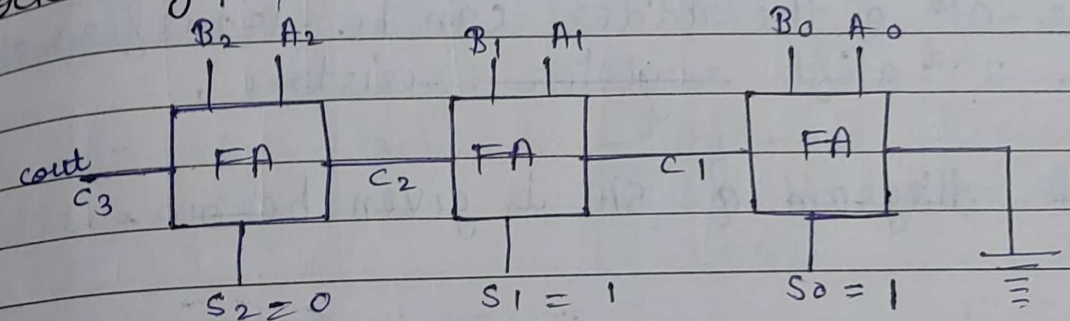
Addend word B — B_3 0 B_2 B_1 B_0

(1) 1 0 0 0

↓

cout

Q. Impliment the binary parallel adder which adds 101 & 110 and then find the op with block diagram.



A = 101 , B = 110

1 0 1

1 1 0

'1' 0 1 1

↓cout

(Truth table)

I/P		O/P	
A	B	S	cout
1	0	1 (S_0)	0
0	1	1 (S_1)	0
1	1	0 (S_2)	1

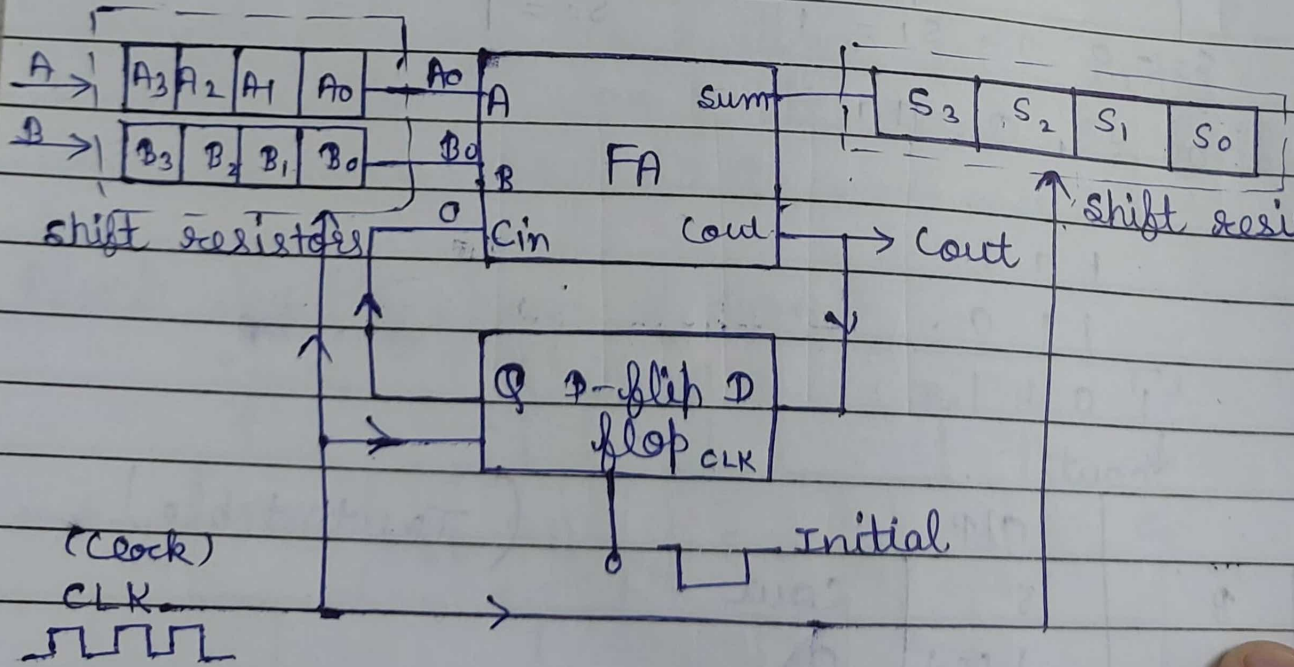
Spiral

So, Sum = 0 11
Cout = 1

Serial adder :-

It is used to add 2 binary no. in serial form. The 2 binary no. to be added serially are stored in 2 ~~st~~ shift register. The ckt adds one pair at a time with the help of 1 full adder. The carry o/p from the full adder is applied to a 'D' flip-flop, the o/p of which is then used as a carry i/p for the next pair of significant bits. The sum bit 'S' from the o/p of the adder can be transferred into the 3rd shift ~~register~~ resistors.

The block diagram of SA is given below.



Page No.
Date

Comparison of parallel and serial adder :-

Parallel	Serial
• It is faster than serial adder.	• It is less fast due to serial addition of bits.
• It requires a no. of full adders equivalent to the no. of bits in binary bits.	• It requires only 1 full adder.
• Basically, it is a combinational CKt.	• It is a sequential CKt.
• All bits are added simultaneously.	• Add. is perform bit by bit starting from LSB.

Magnitude comparator :-

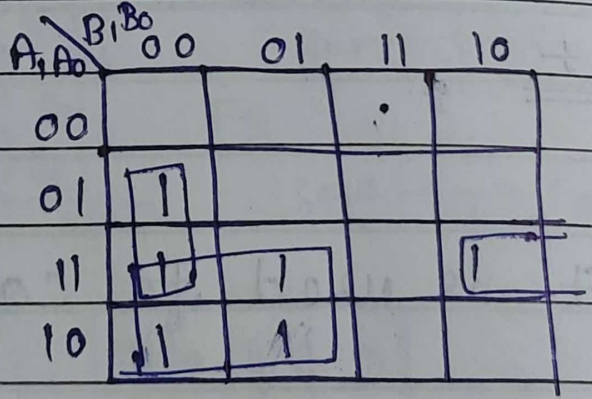
(2) 2-Bit comparator :-

It is used to compare 2-bits of both A & B.

Truth table :-

	A_1	A_0	B_1	B_0	G (A>B)	E (A=B)	L (A<B)
0	0	0	0	0		1	
1	0	0	0	1			1
2	0	0	1	0			1
3	0	0	1	1			1
4	0	1	0	0	1		
5	0	1	0	1		1	
6	0	1	1	0			1
7	0	1	1	1			1
8	1	0	0	0	1		
9	1	0	0	1	1		
10	1	0	1	0		1	
11	1	0	1	1			1
12	1	1	0	0	1		
13	1	1	0	1	1		
14	1	1	1	0	1		
15	1	1	1	1		1	

G :-



$$G = A_1B_1 + \overline{B_1}\,\overline{B_0}A_0 + A_1A_0\overline{B_0}$$

E :-

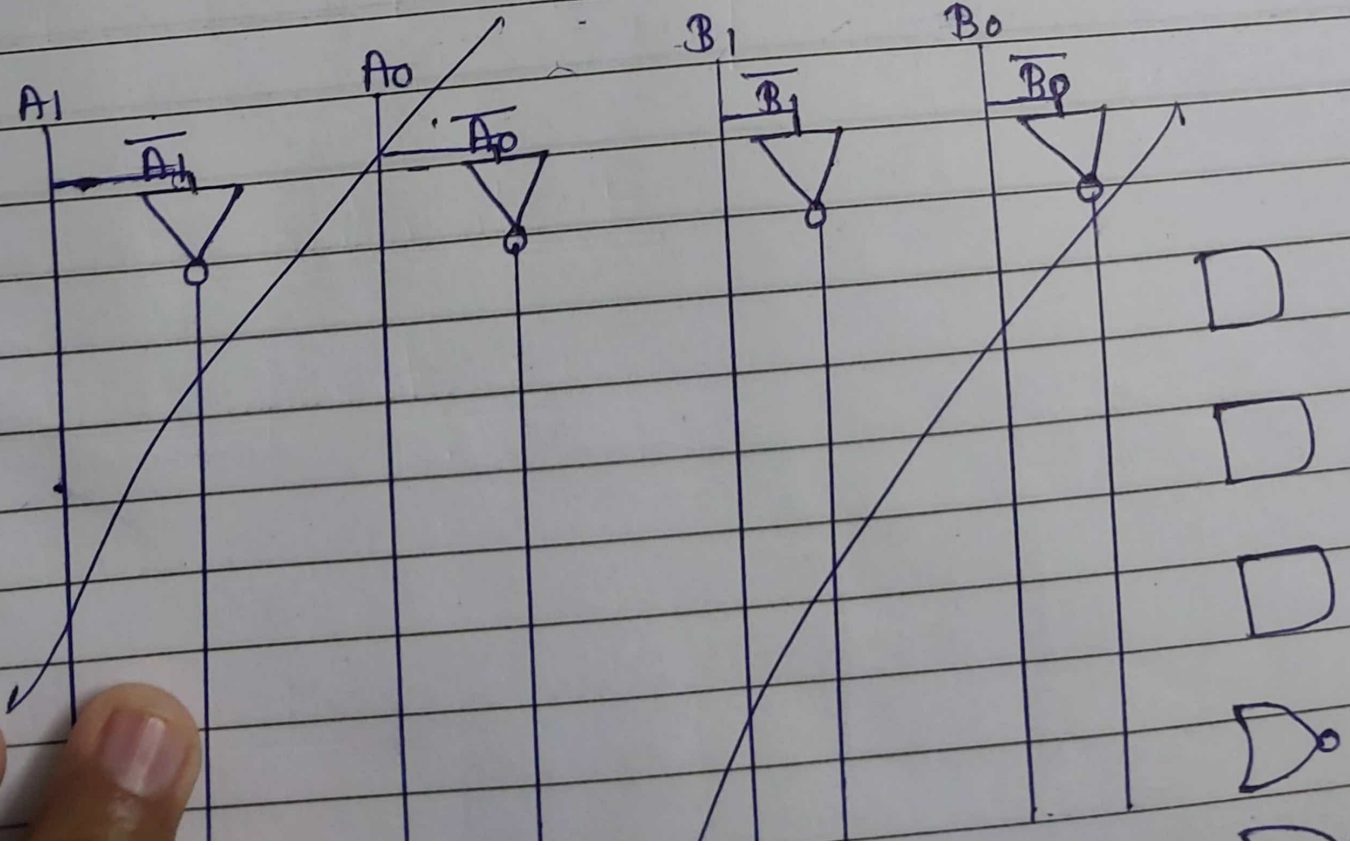
A_1A_0 \ B_1B_0	00	01	11	10
00	1			
01		1		
11			1	
10				1

$$E = \overline{A_1}\,\overline{A_0}\,\overline{B_1}\,\overline{B_0} + \overline{A_1}\,A_0\,\overline{B_1}\,B_0 + A_1A_0\,B_1B_0 + A_1\overline{A_0}\,B_1\overline{B_0}$$

$$E = (A_1 \odot B_1).(A_0 \odot B_0)$$

A_1A_0 \ B_1B_0	00	01	11	10
00		1	1	1
01			1	1
11				
10			1	

$$L = \overline{A_1}\,B_1 + \overline{A_1}\,\overline{A_0}\,B_0 + B_1B_0\,\overline{A_0}$$

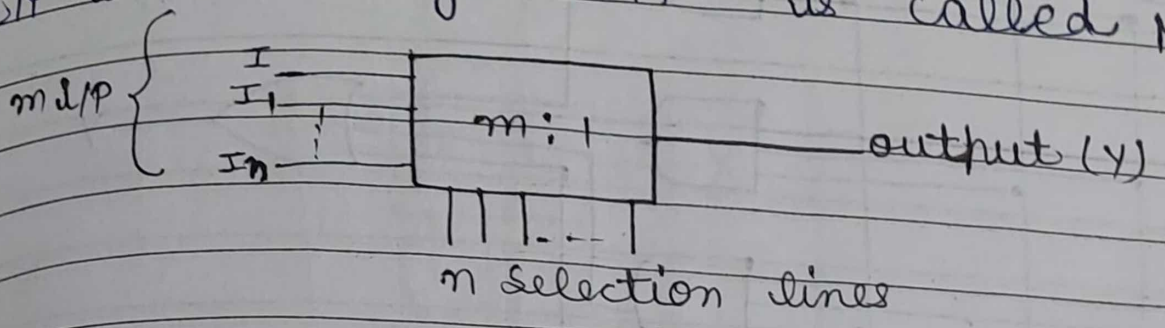


Page No.
Date

Multiplexer :-

The term multiplex means "many into one". It means transmitting large information or i/p to a single o/p.

- The device which is used to convert many i/p into a single o/p is called MUX.



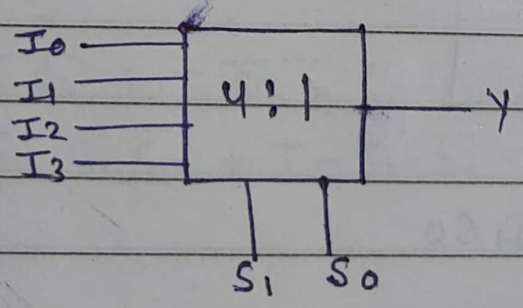
4 : 1 MUX :-

$\rightarrow 2^n = 2^2 \rightarrow$ 2 selection line

Truth table :-

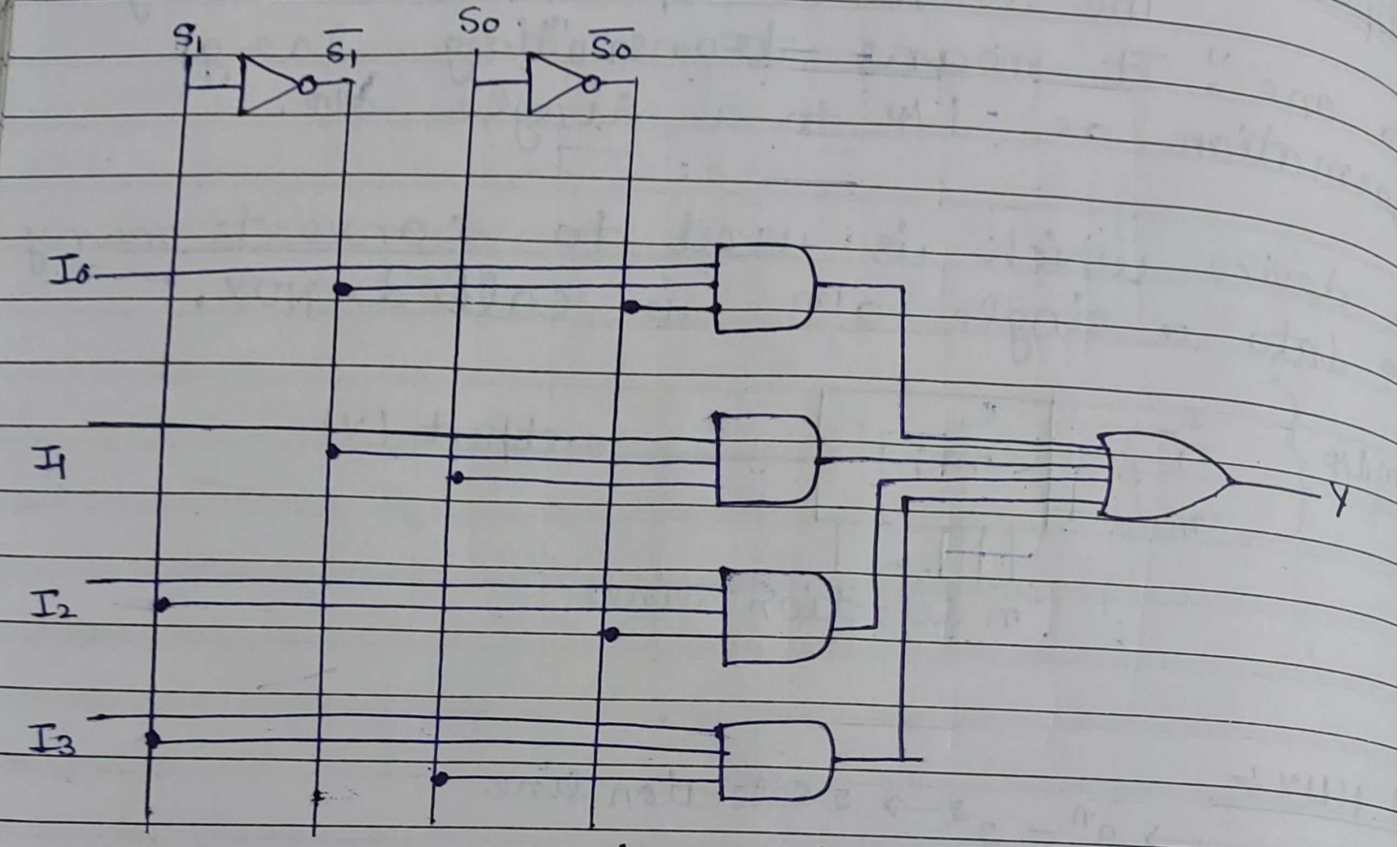
Solution i/p

S_1	S_0	o/p Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

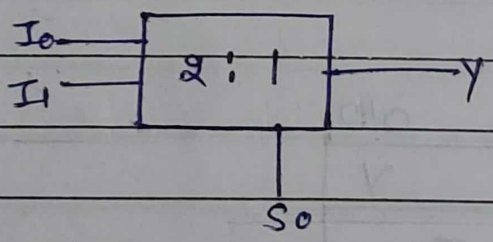


Spiral

$$Y = I_0 \bar{S_1} \bar{S_0} + I_1 \bar{S_1} S_0 + I_2 S_1 \bar{S_0} + I_3 S_1 S_0$$



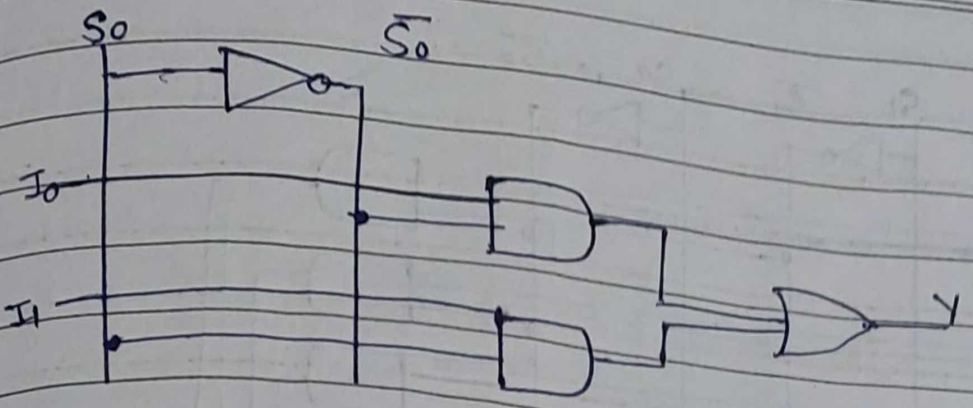
2 : 1



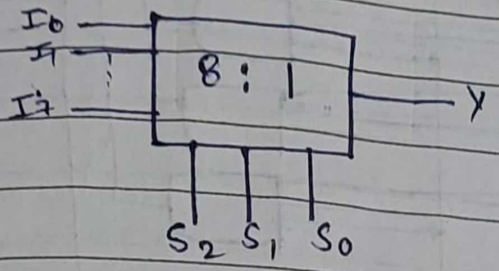
Truth table :-

S_0	Y
0	I_0
1	I_1

$$Y = I_0 \bar{S_0} + I_1 S_0$$



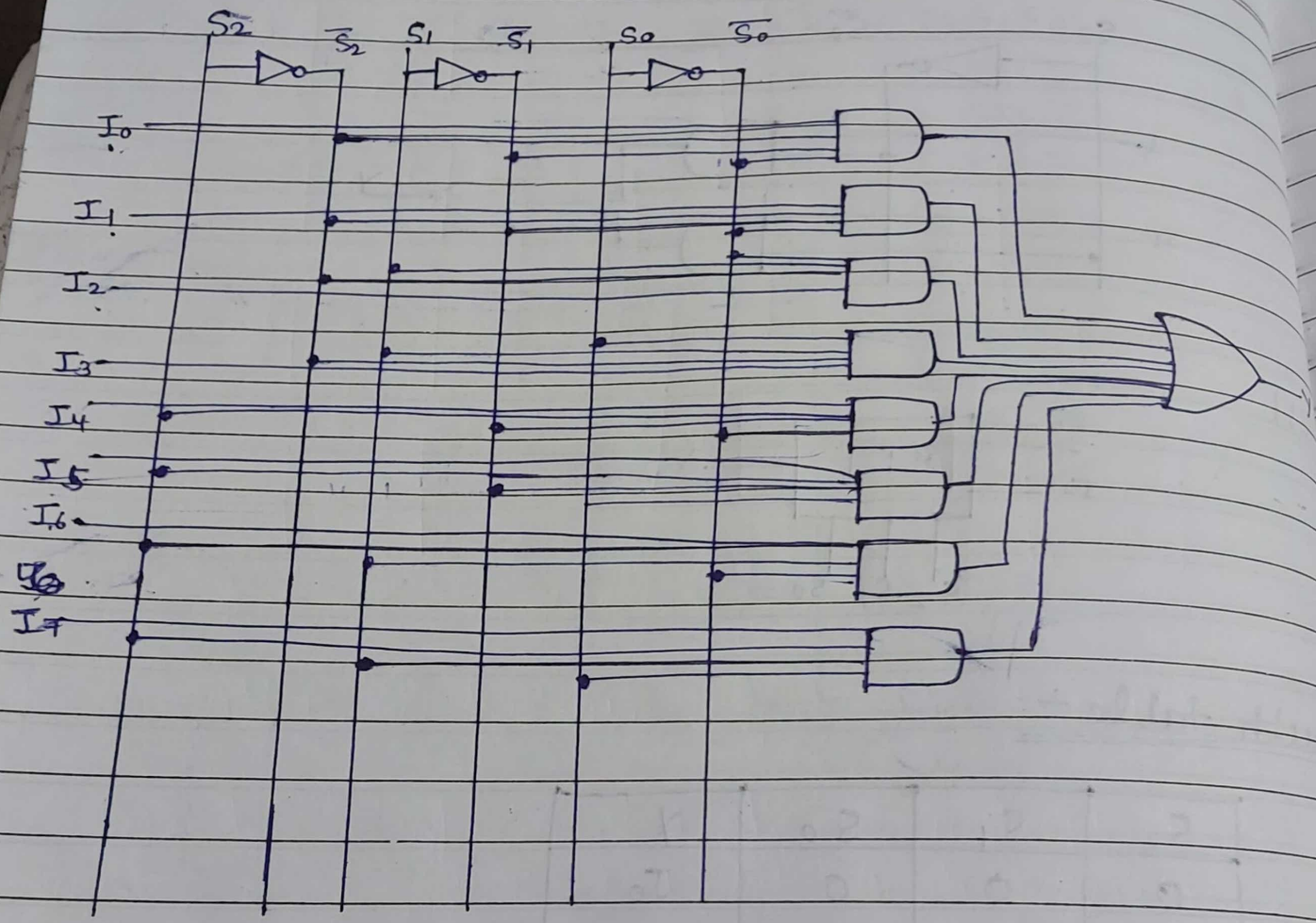
8:1



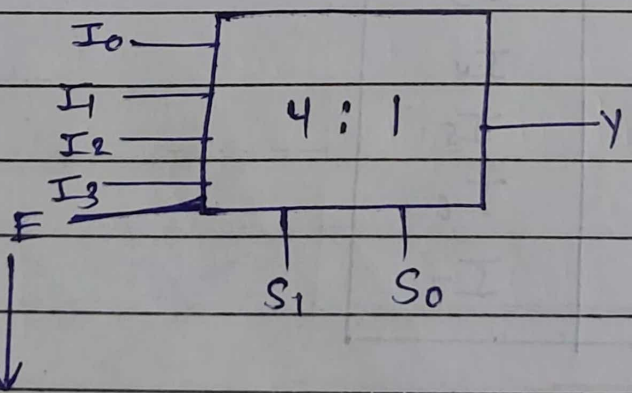
Truth table :-

S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

$$Y = I_0 \overline{S_2}\,\overline{S_1}\,\overline{S_0} + I_1 \overline{S_2}\,\overline{S_1}\,S_0 + I_2 \overline{S_2}\,S_1\,\overline{S_0} + I_3 \overline{S_2}\,S_1\,S_0 + I_4 S_2 \overline{S_1}\,\overline{S_0} + I_5 S_2 \overline{S_1}\,S_0 + I_6 S_2 S_1 \overline{S_0} + I_7 S_2 S_1 S_0$$



MUX with enable i/p :-



Active low $\Rightarrow E = 0$

Active high $\Rightarrow E = 1$

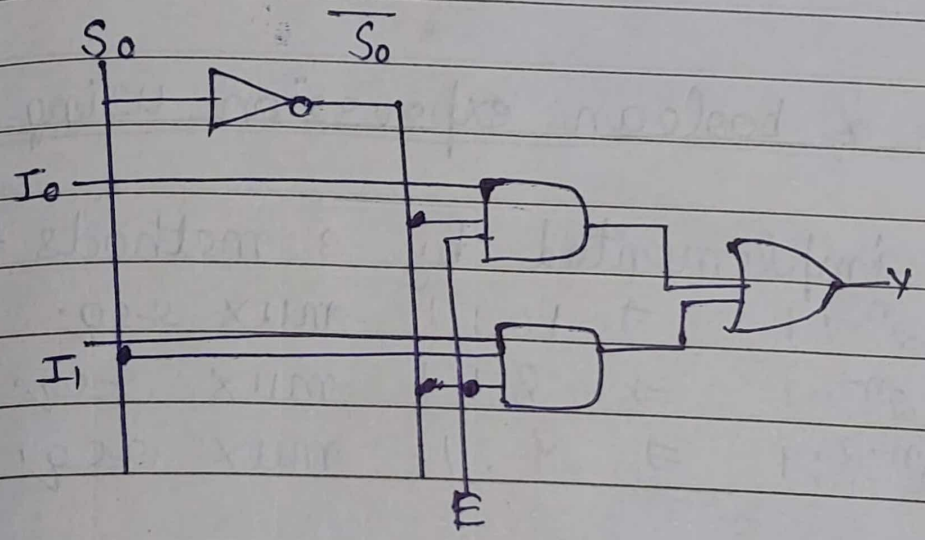
Page No.
Date

Truth table :-

E	S_1	S_0	Y
0	X	X	0
1	0	0	I_0
1	0	1	I_1
1	1	0	I_2
1	1	1	I_3

$Y = I_0 \overline{S_1} \overline{S_0} + I_1 \overline{S_1} S_0 + I_2 S_1 \overline{S_0} + I_3 S_1 S_0$

2:1

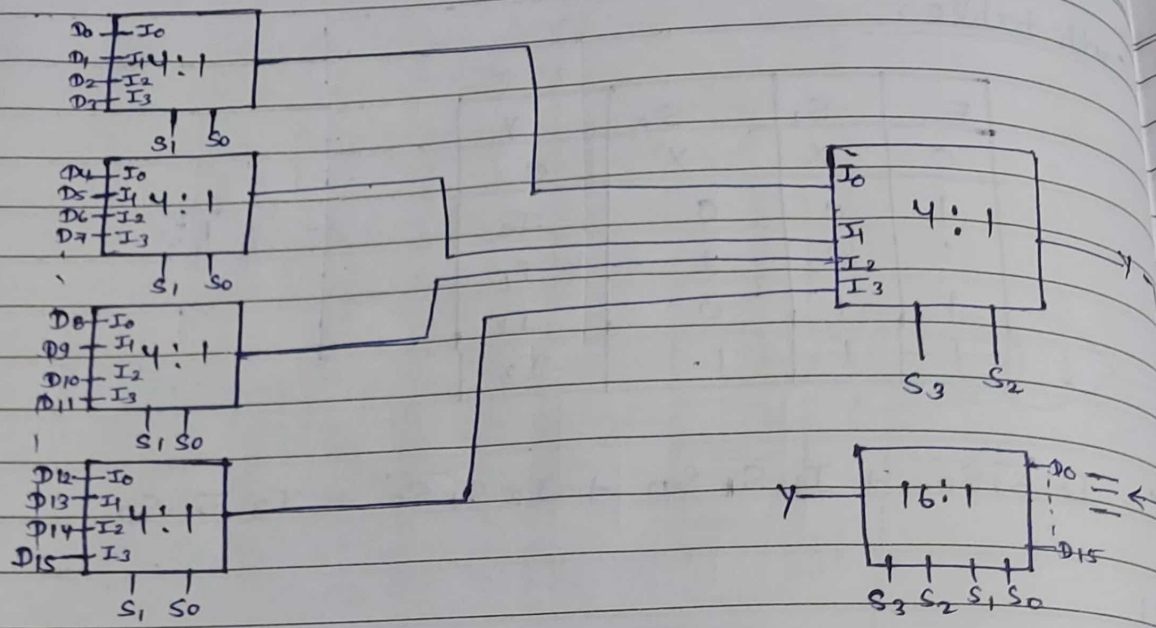


Higher order MUX :-

16:1 MUX :-

2^4 → 4 selection line

By 4:1



Implementation of boolean expression using MUX :-

This can be implemented by 3 methods :-

(1) Type 0 → $2^n : 1$ ⇒ 16 : 1 mux req.

(2) Type 1 → $2^{n-1} : 1$ ⇒ 8 : 1 mux req.

(3) Type 2 → $2^{n-2} : 1$ ⇒ 4 : 1 mux req.

Q. $f(A,B,C,D) = \Sigma m(0,1,2,5,8,13,14)$ implement this boolean expression using type (0,1,2) method

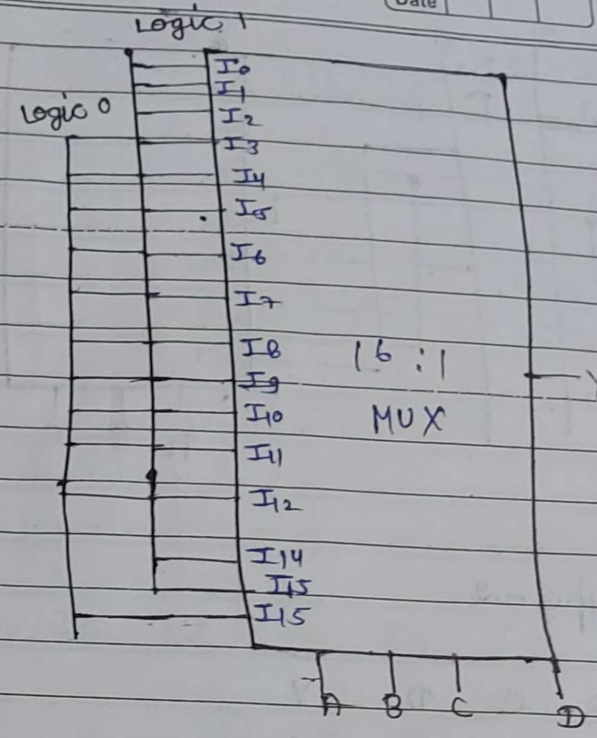
for type 0 – 4 variables

Req. mux = $2^n : 1$

$= 2^4 : 1$

$= 16 : 1$ mux

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0



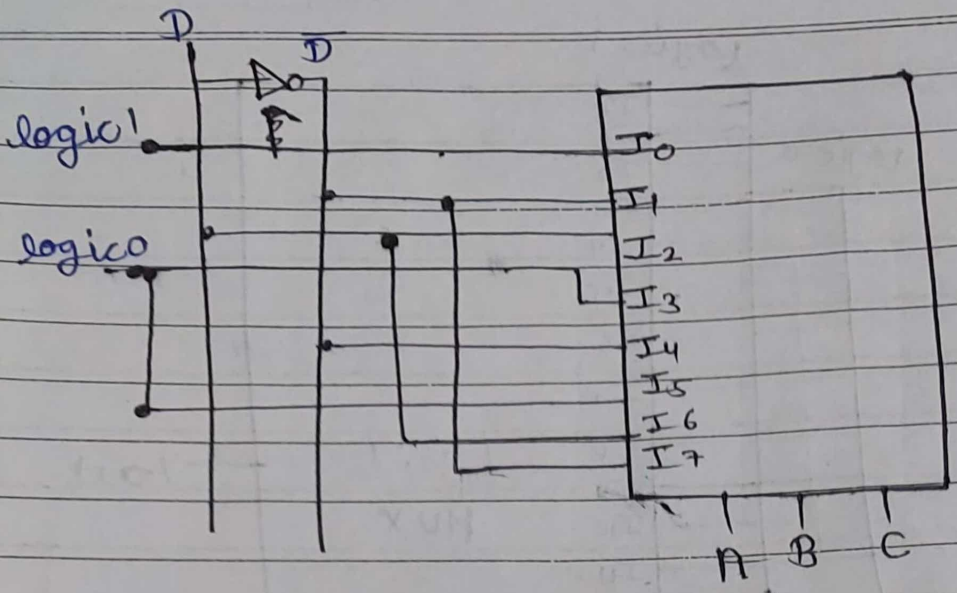
for type -1

Implementation table

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{D}$	0	2	4	6	8	10	12	14
D	1	3	5	7	9	11	13	15
	1	$\bar{D}$	D	0	$\bar{D}$	0	D	$\bar{D}$

A	B	C	Y
0	0	0	1
0	0	1	$\bar{D}$
0	1	0	D
0	1	1	0
1	0	0	$\bar{D}$
1	0	1	0
1	1	0	D
1	1	1	$\bar{D}$

Date



for type-2

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Sp

C \ D	0	1
0	1	1
1	1	0

AB = 00

$I_0 = \bar{C} + \bar{D}$

C \ D	0	1
0	0	1
1	0	1

AB = 01

$I_1 = \bar{C}D$

C \ D	0	1
0	1	0
1	0	0

AB = 10

$I_2 = C\bar{D}$

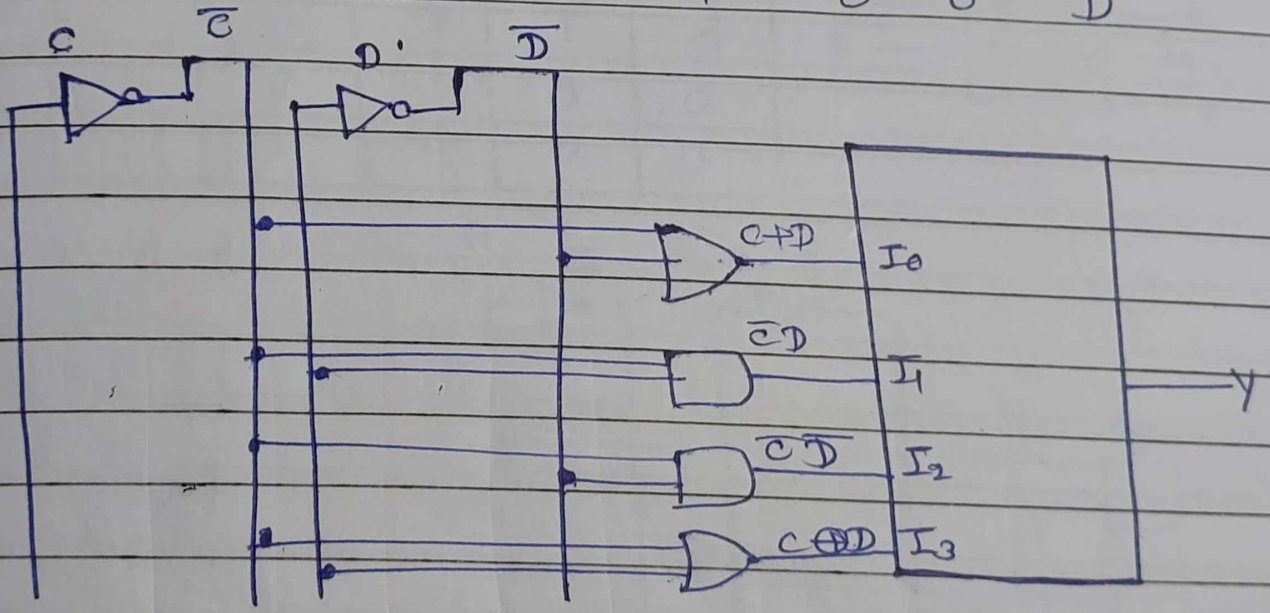
C \ D	0	1
0	0	1
1	1	0

AB = 11

$I_3 = \bar{C}D + C\bar{D}$

Implementation table :-

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
	0	2	4	6	8	10	12	14
D	1	3	5	7	9	11	13	15
$\bar{D}$	1	D	$\bar{D}$	0	1	0	0	D

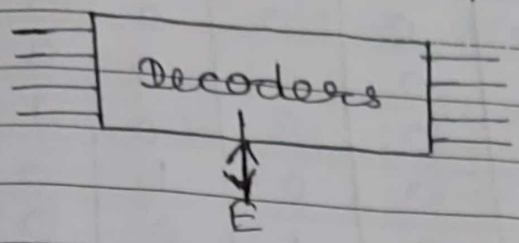


Spiral

Page No.
Date

Decoder :-

It is a multi i/p, multi o/p logic ckt which decodes $n * 2^n$ possible o/p.

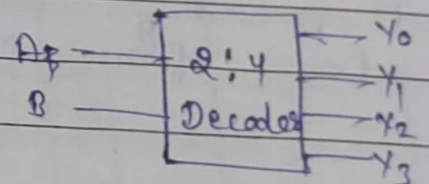


where E=0, decoder is disabled
E=1, decoder is enabled.

ii) 2 to 4 decoder :-

$2^2 = 4$

A	B	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

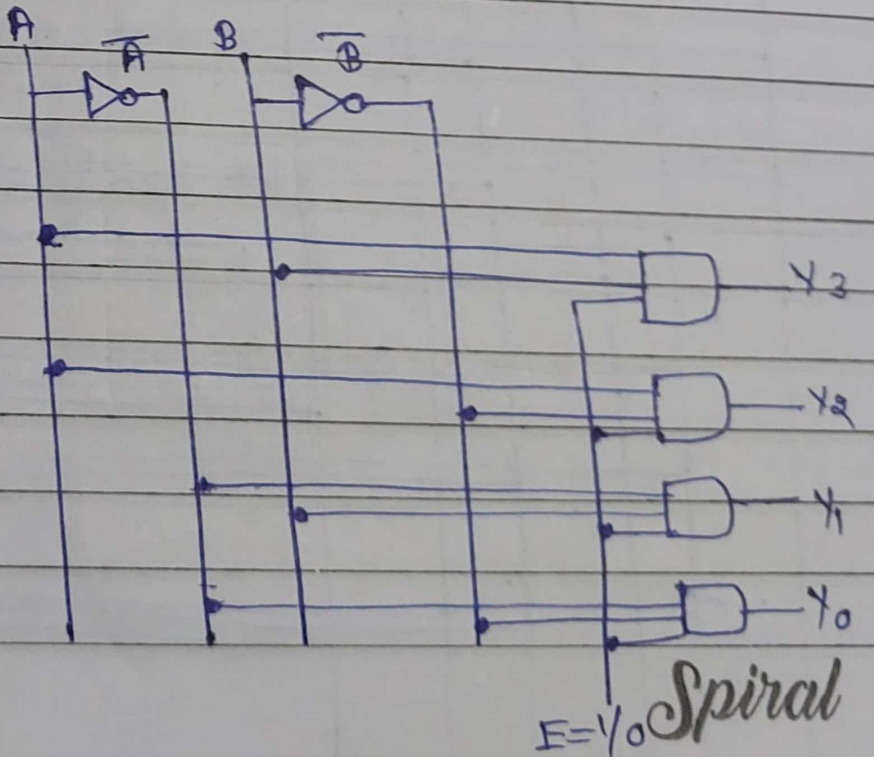


$Y_3 = AB$

$Y_2 = A\overline{B}$

$Y_1 = \overline{A}B$

$Y_0 = \overline{A}\,\overline{B}$

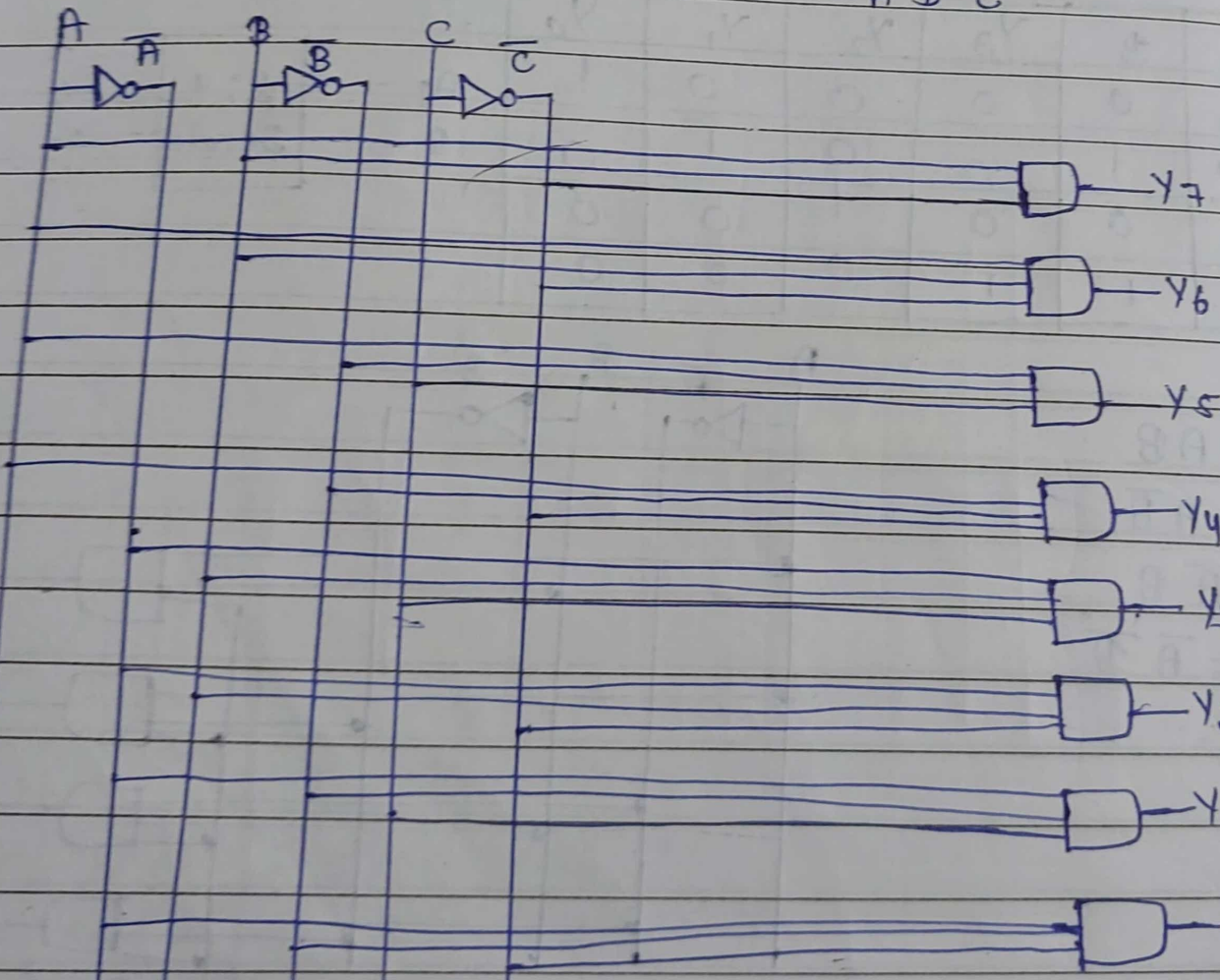


Spiral

(2) 3 to 8 decoder :-

	A	B	C	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	0	1	0
2	0	1	0	0	0	0	0	0	1	0	0
3	0	1	1	0	0	0	0	1	0	0	0
4	1	0	0	0	0	0	1	0	0	0	0
5	1	0	1	0	0	1	0	0	0	0	0
6	1	1	0	0	1	0	0	0	0	0	0
7	1	1	1	1	0	0	0	0	0	0	0

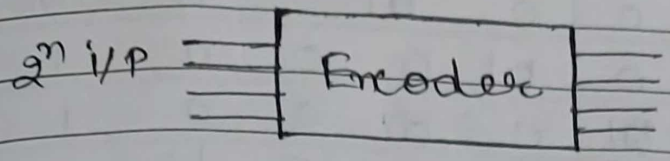
$Y_7 = AB\,C$, $Y_6 = AB\overline{C}$, $Y_5 = A\overline{B}\,C$, $Y_4 = A\overline{B}\,\overline{C}$, $Y_3 = \overline{A}BC$,
$Y_2 = \overline{A}B\overline{C}$, $Y_1 = \overline{A}\,\overline{B}\,C$, $Y_0 = \overline{A}\,\overline{B}\,\overline{C}$



Page No.
Date

Encoder :-

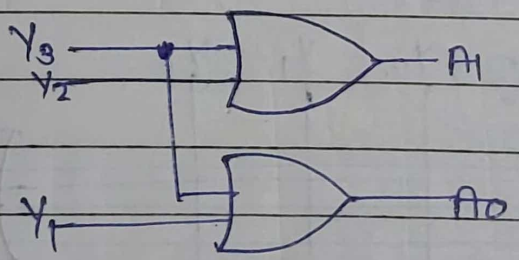
It is a combinational ckt that performs the reverse operation of decoder. It has max. of 2^n i/p lines and n o/p lines.



(i) 4 : 2 encoder :-

	i/p				o/p	
	Y_3	Y_2	Y_1	Y_0	A_1	A_0
0	0	0	0	1	0	0
1	0	0	1	0	0	1
2	0	1	0	0	1	0
3	1	0	0	0	1	1

$A_1 = Y_2 + Y_3$, $A_0 = Y_1 + Y_3$



(ii) 8 : 3 encoder :-

Page No.
Date

Truth table :-

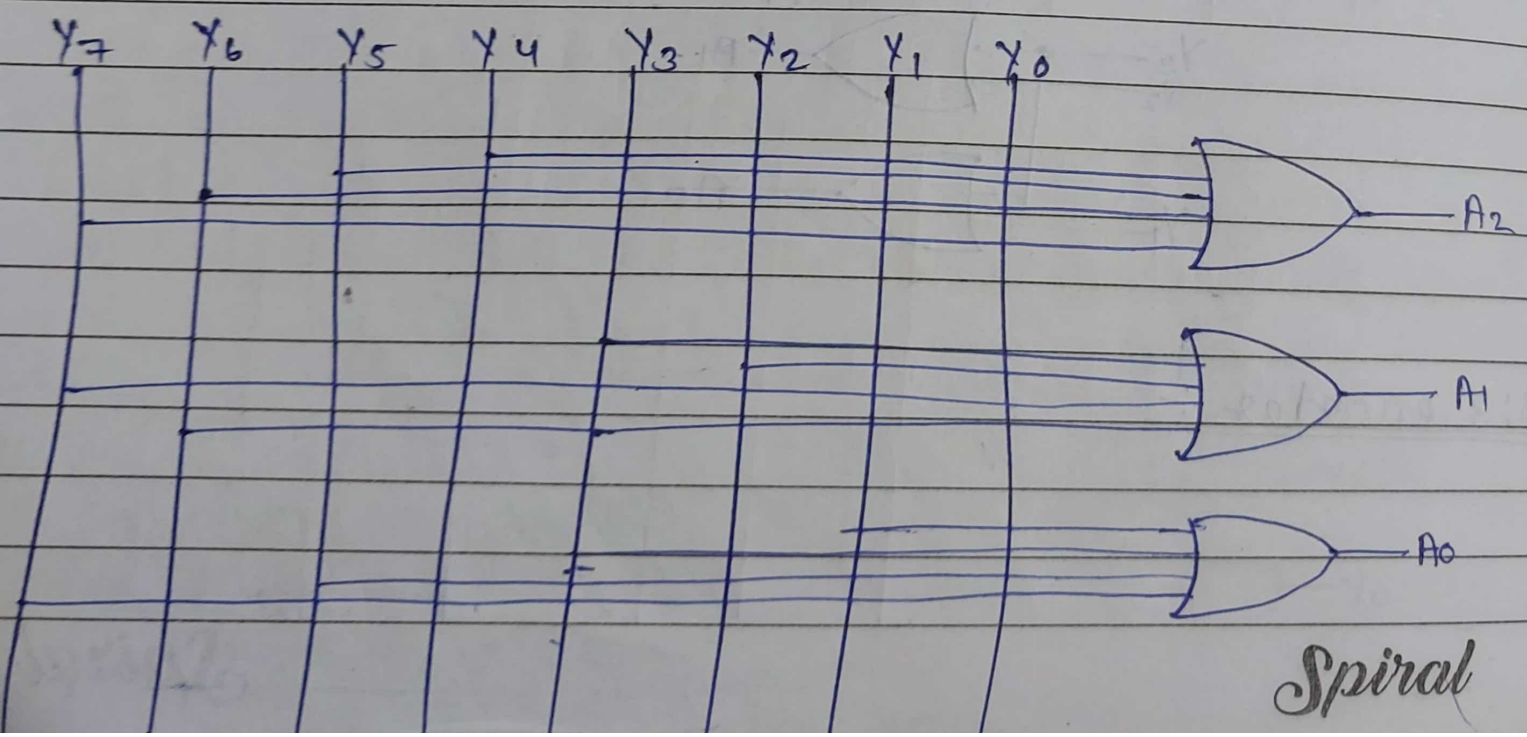
	i/p									
	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0	A_2	A_1	A_0
0	1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	1	0	0	0	1
2	0	0	1	0	0	1	0	0	0	1	0
3	0	0	0	1	1	0	0	0	0	1	1
4	0	0	0	1	0	0	0	0	1	0	0
5	0	0	1	0	0	0	0	0	1	0	1
6	0	1	0	0	0	0	0	0	1	1	0
7	1	0	0	0	0	0	0	0	1	1	1

$A_2 = Y_4 + Y_5 + Y_6 + Y_7$, $A_1 = Y_2 + Y_3 + Y_6 + Y_7$

$A_0 = Y_1 + Y_3 + Y_5 + Y_7$

$A_2 = Y_4 + Y_5 + Y_6 + Y_7$, $A_1 = Y_3 + Y_2 + Y_7 + Y_6$

$A_0 = Y_1 + Y_3 + Y_5 + Y_7$



Spiral